Data Analysis Lecture Quiz

What is the variable type in the following examples:

Gender	Mobile Phone Usage		Total	%
	Not Using	Using		
Male	683	98	781	51.1%
Female	644	102	746	48.9%
Total	1327	200	1527	
%	86.9%	13.1%		

Options: Nominal, ordinal, interval, ratio
Correct answer: Nominal (both)

How many email messages do you receive each day?

1. None (I don't use email)
2. 1-5 per day
3. 6-25 per day
4. 26-100 per day
5. More than 100 per day

Options: Nominal, ordinal, interval, ratio
Correct answer: Ordinal

Please indicate your level of agreement with the following statements.

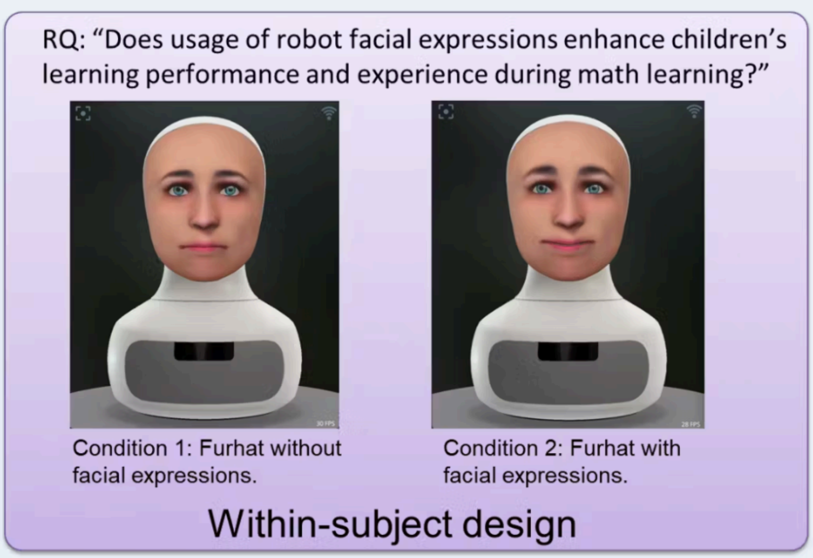
	Strongly disagree	Mildly disagree	Neutral	Mildly agree	Strongly agree
It is safe to talk on a mobile phone while driving.	1	2	3	4	5
It is safe to read a text message on a mobile phone while driving.	1	2	3	4	5
It is safe to compose a text message on a mobile phone while driving.	1	2	3	4	5

Options: Nominal, ordinal, interval, ratio

Correct answer: Interval*

*Since it's hard to define equal intervals for subjective answers, some would argue that this is also ordinal

Which statistical test would you use for the following examples assuming the data is normally distributed:



	Test performance	
Subject ID	Furhat without gestures	Furhat with gestures
1	52	55
2	47	61
3	89	83
4	78	90
5	90	92
6	44	56
7	61	72
8	60	85
9	88	65
10	36	44

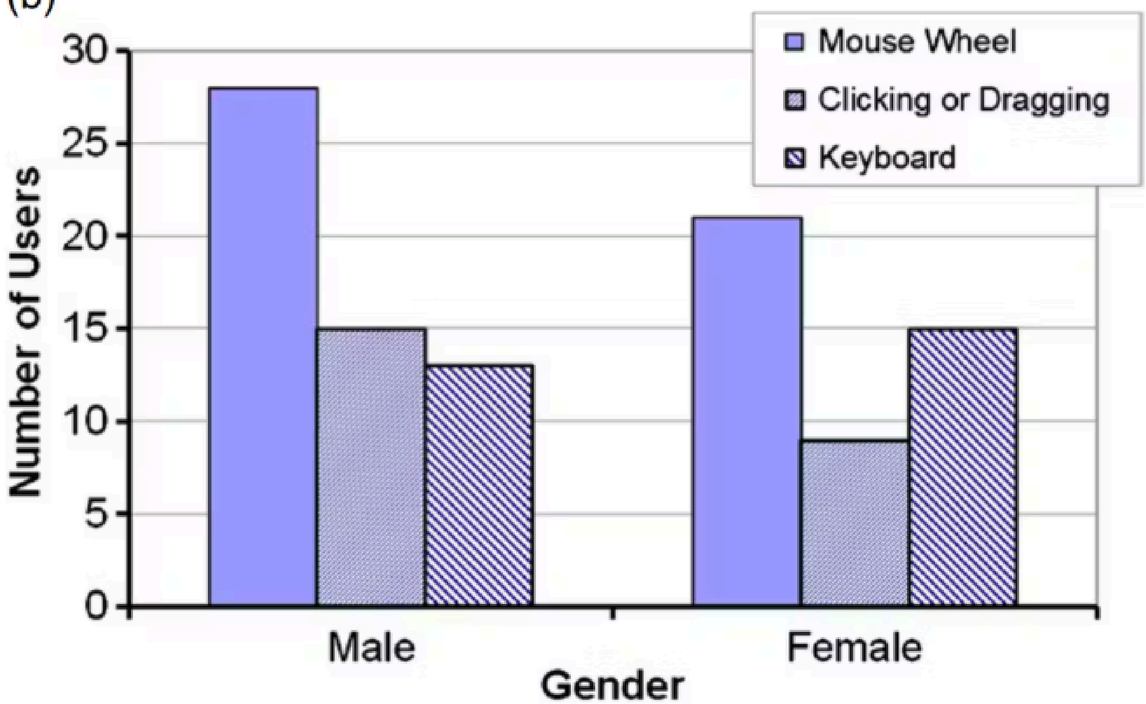
Options: Paired t-test, chi-square test, independent t-test, ANOVA

Correct answer: Paired t-test

(a)

Observed Number of Users				
Gender	Scrolling Method			Total
	MW	CD	KB	
Male	28	15	13	56
Female	21	9	15	45
Total	49	24	28	101

(b)



Options: Paired t-test, chi-square test, independent t-test, ANOVA
Correct answer: Chi-square test

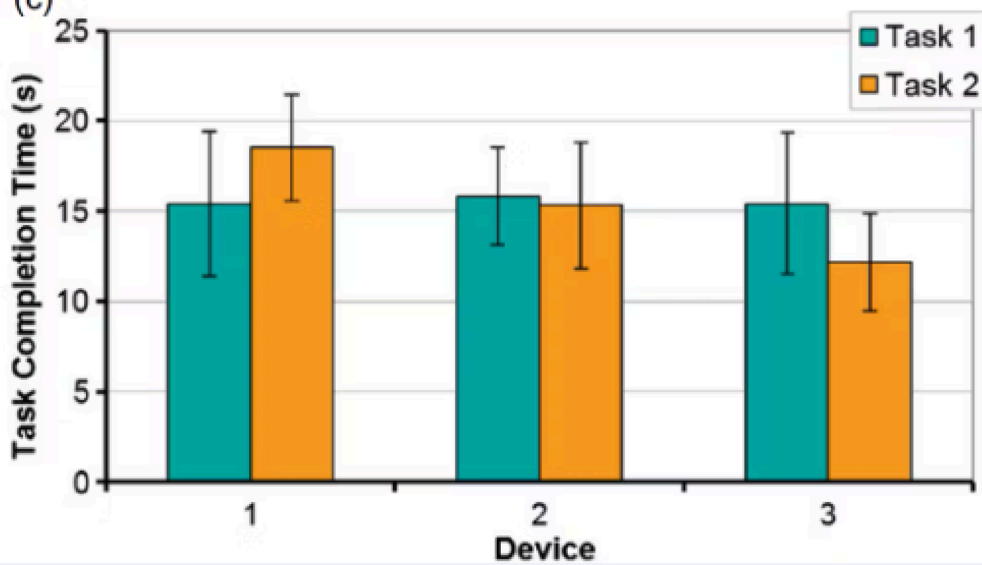
(a)

Participant	Device 1		Device 2		Device 3	
	Task 1	Task 2	Task 1	Task 2	Task 1	Task 2
1	11	18	15	13	20	14
2	10	14	17	15	11	13
3	10	23	13	20	20	16
4	18	18	11	12	11	10
5	20	21	19	14	19	8
6	14	21	20	11	17	13
7	14	16	15	20	16	12
8	20	21	18	20	14	12
9	14	15	13	17	16	14
10	20	15	18	10	11	16
11	14	20	15	16	10	9
12	20	20	16	16	20	9
Mean	15.4	18.5	15.8	15.3	15.4	12.2
SD	4.01	2.94	2.69	3.50	3.92	2.69

(b)

	Task 1	Task 2	*Mean*
Device 1	15.4	18.5	17.0
Device 2	15.8	15.3	15.6
Device 3	15.4	12.2	13.8
Mean	15.6	15.3	15.4

(c)



Options: Paired t-test, chi-square test, independent t-test, ANOVA

Correct answer: ANOVA (two-way ANOVA)